Do Pakistanis Hold Militant Groups in "Low Regard"?*

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Abstract

Blair et al. (2013a) conclude from a 6,000-person endorsement experiment that poor Pakistanis dislike militant groups more than the middle class does, and that Pakistanis in general hold the groups in "low regard." Three of the survey's 75 interviewers recorded the same four answers in most of their interviews, top support for two policies and none for the other two, and in interviews with militant endorsements they recorded one of those answers a notch higher. No other interviewer recorded the first pattern in more than 2% of interviews without endorsements. Their 313 interviews, 5.7% of the total, are almost all with middle- and high-income respondents, and they make those respondents look warmer toward militant groups than the rest of the sample is. Without them, the middle class reacts to a militant endorsement about as negatively as the poor do. The gap between the two, Blair et al.'s central estimate, falls from 2.2 to 1.0 points on a 0–100 scale and is not distinguishable from zero in any of their specifications; the larger contrasts they report among the urban poor and in violent districts shrink by a fifth to a third. The evidence for low regard is a small average effect, and it stays small: a militant endorsement costs a policy 2.1 points without the three interviewers, against 1.3 with them. The standard measurement model's estimate of the share who react favorably moves from 43% to 38%, but with a mean reaction this close to zero the share follows from the model's assumption that reactions are normally distributed, and a population in which no one reacts favorably fits the data as well. The three interviewers also produce the published data's appearance that endorsements push respondents out of the middle of the scale in both directions, the pattern offsetting reactions would produce; without them it disappears.

1 Introduction

Do Pakistanis hold militant groups in "low regard"? Blair et al. (2013a) fielded a 6,000-person endorsement experiment to study how poverty relates to support for four militant groups, using a design that sidesteps the nonresponse and social desirability problems of direct questions. Their central finding is that poor Pakistanis dislike militant groups more than middle-class Pakistanis do: in their data, the middle class does not react to a militant endorsement at all, and the poor react against it. Along the way they report that an endorsement by any of the four groups lowers support for a policy, and they conclude that Pakistanis in general are weakly negative toward these organizations. The effect is small. In their replication data, support falls by 1.3 points on a 0–100

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scale when a militant group is said to back the policy. Being associated with al-Qa'ida or the Afghan Taliban appears to cost a policy almost nothing.

Both conclusions rest in part on three interviewers. In most of their interviews without endorsements, interviewers 7, 37, and 71 recorded the same four answers: top support for polio vaccination and curriculum reform, none for Frontier Crimes Regulation reform or the Durand Line jirga. In most of their interviews with endorsements, they recorded the same answers with Frontier Crimes Regulation support one notch higher. No other interviewer of the survey's 75 recorded either combination often (Figure 1). The three recorded 313 interviews, 5.7% of the total, and 99% of their respondents with earnings data are middle or high income by Blair et al.'s classification. Because their endorsed interviews show more support than their unendorsed ones, they make militant endorsements look less costly, and less costly to the middle class in particular.

Without them, Blair et al.'s central estimate does not hold. They compare the effect of an endorsement among respondents in the bottom fifth of household earnings with the effect among the middle three fifths. The poor's reaction is the same with and without the three interviewers. The middle class's, indistinguishable from zero with all interviews, is a cost of 1.7 points without them, against the poor's 2.6. The gap between the two falls from 2.2 to 1.0 points, and in none of Blair et al.'s specifications does it remain more than 1.5 standard errors from zero. Dropping three interviewers chosen at random moves the gap that far in 1 of 1,000 draws. The larger contrasts they report among the urban poor and in violent districts shrink by a fifth to a third and remain, and their finding that district-level poverty does not change the effect is unchanged (Section 5).

The evidence for low regard is the small average effect, and it stays small. Without the three interviewers, a militant endorsement costs a policy 2.1 points rather than 1.3. Under the measurement model of Bullock et al. (2011), the estimated probability that a respondent reacts favorably to an endorsement moves from 43% (95% credible interval 31%–52%) with their interviews to 38% (29%–45%) without them, a change much smaller than the uncertainty in either estimate. But that probability comes from the model's form. The model gives each group's reactions a normal distribution, and when the mean reaction is close to zero, as it is here, any spread at all places a large share of that distribution above zero. The data cannot tell such a population from one in which everyone is mildly opposed or indifferent.

The three interviewers also bear on whether the small average hides large reactions in both directions. A small average has three readings. Most respondents could dislike the groups mildly; most could be indifferent; or large numbers could like and dislike the groups, with reactions that nearly cancel. Because each respondent answers four policy questions, the design can separate them: a respondent who reacts to militant endorsements reacts on all four, which moves the average of their four answers well away from where it would otherwise be. With all interviews, respondents whose answers average the middle of the scale become 3.5 percentage points rarer under endorsement, and the positions just above and just below both gain, as large offsetting reactions would produce. Once standard errors account for interviewers, even that drop is not distinguishable from zero. The three interviewers' answers average exactly the middle of the scale without endorsements and just above it with them, and they account for all of the movement out of the middle and most of the movement above it. Without them, the share at the middle no longer falls and the position just

above no longer gains. Simulations built from respondents who saw no endorsements show that opposition alone fits what remains, and so do simulations in which up to 30% react favorably.

How many respondents react favorably is therefore not identified by the data. The measurement model's share comes from its assumption that reactions are normally distributed around a small negative mean, and nothing model-free in these data confirms or rules out a favorable minority of that size.

2 Endorsement Experiments

Endorsement experiments randomize whether a policy question mentions that an actor of interest supports the policy. Respondents in the control condition evaluate the policy alone; those in the treatment condition evaluate it with the endorsement. The difference in average support identifies the effect of the cue on stated support. Reading that effect as affect toward the endorser requires that respondents understand the cue and that it works through their attitudes toward the group. The design has been used to study militant groups in Pakistan and combatants in Afghanistan (Bullock et al., 2011; Blair et al., 2013a; Lyall et al., 2013; Blair et al., 2014).

A small negative average treatment effect can reflect many small negative reactions, a few large ones, or positive and negative reactions that nearly cancel. Suppose a proportion p of respondents reacts favorably and the rest unfavorably, with average effects δ^+ and δ^- on the observed response. Then

$$\beta = p\delta^+ + (1-p)\delta^-. \tag{1}$$

One observed mean cannot determine three unknowns (Manski, 1990; Lu et al., 2018), and endorsements by divisive actors do move different respondents in opposite directions (Nicholson, 2012). Mild disapproval (p near zero, small negative δ^-), indifference (δ^+ and δ^- near zero), and offsetting reactions (large δ^+ and δ^- of opposite sign) can all produce the same β. When each respondent answers several questions, the joint distribution of their answers carries more information: a reaction carried across questions moves a respondent's average answer further than independent noise on each answer does.

3 The Experiment and Data

Blair et al. (2013a) surveyed 6,000 adults in Pakistan's four main provinces in 2009. Respondents rated polio vaccination, Frontier Crimes Regulation reform, the Durand Line jirga, and madrassa curriculum reform on a five-point scale from "not at all" to "a great deal." Respondents in the endorsement condition heard a different group endorse each policy: Kashmiri tanzeems, the Afghan Taliban, al-Qa'ida, and sectarian tanzeems. I rescale answers to $[0, 1]$, so that each step on the scale is 0.25; a mean on this scale is not a percentage of respondents supporting a policy.

I use Blair et al.'s replication file (Blair et al., 2013b), which records each respondent's interviewer, district, and province. The analysis sample has 5,479 respondents who gave a substantive answer to all four policies and have a recorded interviewer; 35 of them received endorsements

on some policies but not others and are left out of comparisons between respondents in the two versions: the interviewer effects and the within-respondent analyses. The `endorse` R package distributes a version of the same data without interviewer identifiers (Shiraito and Imai, 2022). Except in Section 5, which reproduces Blair et al.'s estimates with their weights and sample, the analysis is unweighted. Blair et al. use population and policy weights, and their survey oversampled Balochistan and Khyber Pakhtunkhwa, so the unweighted estimates describe the sample rather than Pakistan.

Endorsement versions were assigned within interviewers, and the 75 interviewers recorded between 6 and 285 interviews each (median 52). Answers recorded by the same interviewer tend to resemble one another more than answers recorded by different interviewers (West and Blom, 2017), and when the effect itself varies across interviewers, inference about it must treat interviewers as the units sampled even though versions were randomized within them (Abadie et al., 2023). Standard errors for the average effect are therefore clustered by interviewer, and every bootstrap interval resamples interviewers, keeping each interviewer's respondents in both conditions together.

4 Three Interviewers

Every interviewer conducted interviews in both versions, so each yields an estimate of the treatment effect. If interviewers differed only by sampling noise, those estimates would scatter around the pooled effect by no more than their standard errors allow. They do not. Among the 60 interviewers with at least 20 respondents and some variation in recorded answers, and leaving out the 35 respondents recorded with both versions, Cochran's Q, the inverse-variance weighted sum of squared deviations of the interviewer effects from their weighted mean, is 118.0. Versions were randomized within interviewers, so reshuffling each interviewer's version labels, after removing the pooled effect, shows how large Q gets when interviewers differ only by chance, a randomization test of effect variation across strata (Ding et al., 2016). Against 19,999 reshuffles, $p < 0.001$. The usual chi-square reference with 59 degrees of freedom gives $p < 0.001$ too, but it understates how much Q varies when each interviewer's standard error comes from a few dozen respondents, so the reshuffles are the reference used throughout. Keeping the respondents recorded with both versions lowers Q to 93.1 ($p = 0.020$ from reshuffles). Almost all of that difference comes from one of them, interviewed by interviewer 71 and recorded as 4-1-0-4 with polio vaccination unendorsed. Interviewer 71's other answers vary so little that this one interview raises their standard error by 37.1%. Table 1 lists the interviewers furthest from the pooled effect. Interviewers 71, 37, and 7 rank 1, 2, 3, with effects of +3.7 to +8.2 points: in their interviews, an endorsement by a militant group raises support. Leaving one interviewer out at a time, removing interviewer 37 moves the pooled effect more than removing any other (the three rank 1, 3, 7). Without the three, the interviewer effects are consistent with sampling noise ($Q = 52.6$, $p = 0.753$ from reshuffles). Another 2 interviewers recorded identical answers on the four policies in every interview, so their effects have no sampling variance at all.

Figure 1 shows the 75 interviewers four ways. Panel A shows how many interviews each recorded; by number of interviews the three interviewers at issue rank 12, 19, 37 of 75, so none stands out by volume. Panel B compares how often each interviewer recorded their single most common

Table 1: Interviewer-specific treatment effects furthest from the pooled effect

Interviewer	Respondents	Effect (points)	SE	z	Shift when left out
71	51	+5.7	1.0	+7.1	−0.2
37	160	+8.2	2.4	+4.0	−0.5
7	101	+3.7	2.1	+2.3	−0.1
34	46	+2.7	1.7	+2.3	0.0
66	174	−7.3	2.8	−2.1	+0.2
54	63	−11.3	4.9	−2.1	+0.1
51	24	−10.8	5.3	−1.8	0.0
16	41	−8.6	4.1	−1.8	+0.1

All interviews except the 35 respondents recorded with both versions. Effect is the difference between mean answers with and without endorsements among an interviewer's respondents, in points on a 0–100 scale, with a standard error clustered by respondent. z compares the effect with the pooled effect, −1.3 points. Shift is the change in the pooled effect, in points, when the interviewer's respondents are removed. Interviewers with fewer than 20 respondents or no variation in recorded answers are omitted.

combination of four answers with how often respondents in the same districts give it, leaving aside top support on all four policies, the most common combination in the survey. Many interviewers repeat a combination more often than their districts do, and 16 do so more often than in 99.9% of samples drawn from their districts' respondents. Interviewers 7, 37, and 71 are among them but are not unique on this measure.

What sets the three apart is which combination they repeat and how it changes with the questionnaire version (panel D). In their interviews without endorsements they recorded top support for polio vaccination and curriculum reform and no support for the other two policies, 4-0-0-4 on a 0–4 scale, in 58% to 94% of interviews. In their interviews with endorsements they recorded 4-1-0-4 in 55% to 65%. No other interviewer recorded 4-0-0-4 in more than 2% of interviews without endorsements or 4-1-0-4 in more than 15% of interviews with them. Their interviews also split unevenly between versions (panel C): 187 without endorsements and 125 with them, more uneven than random assignment at the survey's overall rate would produce (two-sided binomial $p = 0.001$).

Two tests put a probability on these patterns. The first asks about the block directly. Of the 446 interviews without endorsements that other interviewers recorded in the districts where the three worked, none were 4-0-0-4; across the whole survey, other interviewers recorded it in 0.1% of such interviews. Suppose each of the three interviewers' respondents gave 4-0-0-4 without endorsements, or 4-1-0-4 with them, at the rate respondents of other interviewers in the same districts do, with one match added to each comparison group so that no rate is zero. The chance of recording as many as each of the three did is below 10^{-12} for every interviewer and version. The block was found by looking at the data, so the test should allow for everything that could have been found: the 93,750 combinations of answer pattern, version, and interviewer. Multiplying by that number leaves every probability below 10^{-7}. The test treats respondents as independent. Respondents who agree with their neighbors would make it overstate the evidence, but the neighbors here, other interviewers' respondents in the same districts, did not give the block at all.

The second test names no answer pattern and assumes nothing about how respondents resemble one another. For each interviewer, it measures how different the answers in the two versions are: the

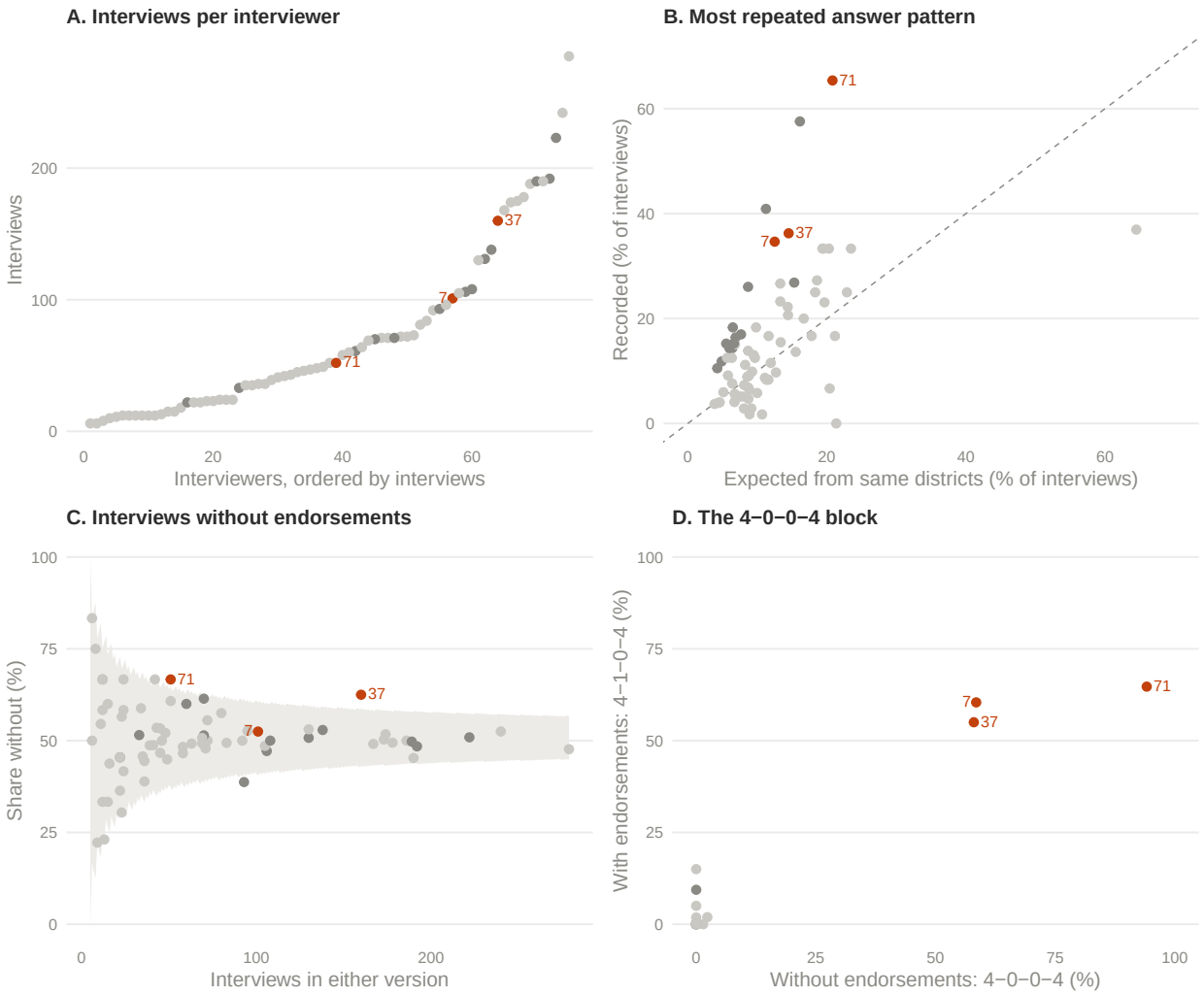


Figure 1: The survey's 75 interviewers. Interviewers 7, 37, and 71 in orange; other interviewers flagged by the repetition screen in dark gray. A: interviews per interviewer. B: share of an interviewer's interviews taken by their most common combination of four answers, excluding top support on all four policies, against the mean share in samples of the same size drawn from respondents in the same districts; the dashed line marks equality. C: share of an interviewer's interviews without endorsements, with the band containing 95% of shares under random assignment at the survey's overall rate. D: share of interviews without endorsements recorded as 4-0-0-4 against share of interviews with endorsements recorded as 4-1-0-4.

share of interviews that would have to change their combination of four answers for the two versions to match. Reshuffling the interviewer's version labels 9,999 times shows how large that share gets by chance. Figure 2 shows the result for all 75 interviewers. For interviewers 7, 37, and 71, no reshuffle produced a share as large as the observed one; allowing for having tested 75 interviewers, $p \leq 0.008$. Interviewers 2 and 61 also fall outside every reshuffle, but in a different way: their answers are spread over many combinations (none takes more than 8% of their interviews, against at least 35% for each of the three), and they record less support with endorsements, not more. The test also responds to ordinary endorsement effects, and 17% of the remaining interviewers have $p < 0.05$. What it shows about the three is that their answers change with the version far more than chance allows; that the change runs toward more support is what Table 1 shows.

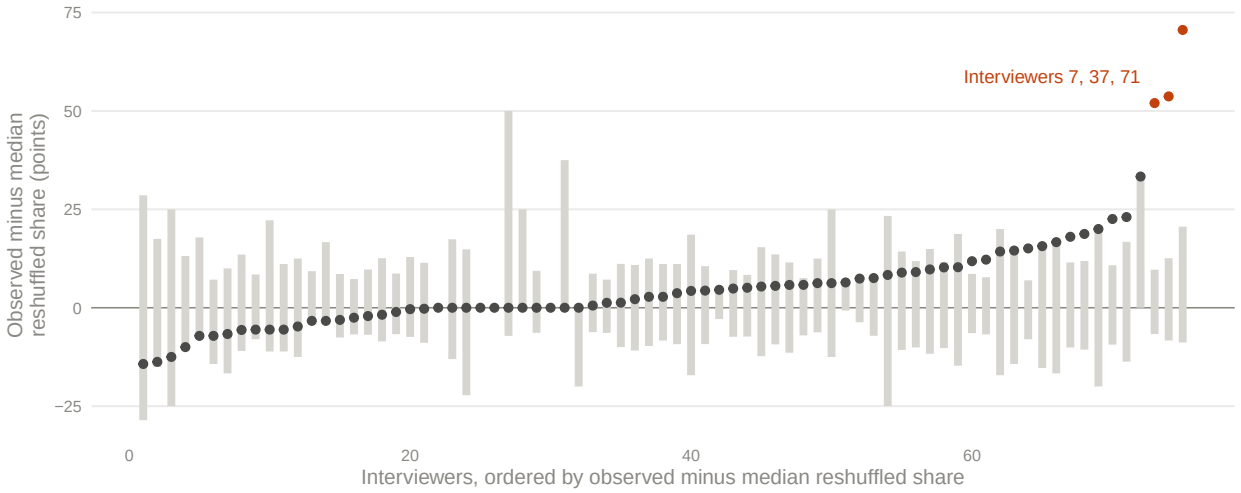


Figure 2: How different each interviewer's interviews are in the two versions. For each interviewer, the share of interviews that would have to change their combination of four answers for the two versions to match, minus the median share from 9,999 reshuffles of that interviewer's version labels. Gray bars contain the middle 95% of reshuffled shares, on the same scale. Interviewers 7, 37, and 71 in orange.

Whether these interviewers filled in forms themselves, coached respondents, or recorded answers in some other way cannot be told from the data. Duplicated answer strings are a known sign of fabricated interviews (Kuriakose and Robbins, 2016), but the question here is narrower. Answers this uniform do not behave like independent responses, and they produce exactly the features of the data that the reading of offsetting reactions rests on.

The 4-0-0-4 combination averages 8/16, the middle of the scale, and 4-1-0-4 averages 9/16, one step above it. Of the 291 respondents without endorsements whose answers average the middle of the scale, 122 were interviewed by the three; of the 187 with endorsements, 8 were. Their shift from one combination to the other moves respondents out of the middle and into the position just above it, and it makes their endorsed interviews record more support than their unendorsed ones. Table 2 shows the consequences. With all interviews, the share of respondents averaging the middle falls by 3.5 points under endorsement. With the interviewer-clustered interval, even that change is not distinguishable from zero (-3.5 [-8.8, $+0.7$]). Without the three interviewers, the share at the middle barely changes and the gain one step above disappears, while the gain one step below

remains. The average effect grows from −0.013 to −0.021. Dropping every interviewer flagged by the repetition screen, or the interviewers with duplicated interviews (Section 6), leaves the changes around the middle much as they are without the three, and the average effect grows further, to −0.028 and −0.023.

Table 2: The average effect and the middle of the scale, with and without the three interviewers

Sample	Respondents	Interviewers	Average effect	Midpoint (pp)	One above (pp)	One below (pp)
All interviews	5,479	75	−0.013	−3.5	+2.7	+2.0
			(0.007)	[−8.8, +0.7]	[+0.2, +5.6]	[+1.0, +3.2]
Without 7, 37, 71	5,166	72	−0.021	+0.5	+0.5	+2.2
			(0.005)	[−1.1, +1.8]	[−0.7, +1.9]	[+1.1, +3.4]
Without flagged	3,728	59	−0.028	+0.7	+1.2	+2.9
			(0.006)	[−1.1, +2.2]	[−0.5, +2.8]	[+1.7, +4.3]
Without 7, 37, 71, duplicating	3,861	59	−0.023	+0.9	+0.5	+2.0
			(0.006)	[−0.8, +2.5]	[−1.0, +2.1]	[+0.9, +3.3]

Average effect is the difference between mean answers with and without endorsements on the 0–1 scale, with interviewer-clustered standard errors in parentheses below. Midpoint, one above, and one below are changes, in percentage points of respondents, in the shares whose four answers average 8/16, 9/16, and 7/16; brackets below are 95% bootstrap intervals resampling interviewers. Flagged interviewers repeat one combination of answers more often than in 99.9% of samples from respondents in their districts. Duplicating interviewers recorded at least one pair of interviews identical on every question (Section 6).

5 Poverty and Support without the Three

Blair et al.'s main result is a difference between treatment effects. They classify respondents by monthly household cash earnings within province and urban or rural residence: the bottom fifth as low income, the top fifth as high income, the rest as middle income. Their outcome is the mean of a respondent's four answers divided by five, so it runs from 0.2 to 1 and one step on the answer scale is 0.2; multiplying their estimates by 125 gives points on the 0–100 scale used elsewhere here. A weighted regression with sampling points as clusters and provinces as strata estimates the effect of an endorsement in each income group and the differences between groups. In the first column of their Table 2, the effect for the middle class is −0.001 (standard error 0.006), the effect for the poor is −0.019, and the difference, −0.018 (0.010), is their central estimate. Later columns add demographic and attitudinal controls and interactions of the cue with every control.

I rebuilt their analysis file and estimates from the Stata code in their replication archive. The sample sizes and every coefficient and standard error in their Tables 1 through 5 match the published values to the third decimal, except one.[1] Table 3 reports the quantities that carry their conclusions, with all interviews and without interviewers 7, 37, and 71.

Without the three, the poor react to endorsements as before, −0.021 against −0.019, and the middle class reacts nearly as much: −0.013 (0.005) against −0.001. The gap between them falls from −0.018 to −0.008 (0.010). With Blair et al.'s demographic and attitudinal controls it falls

[1]The interaction of the cue with low district income in their Table 3 is −0.016 here and −0.015 in print, with standard errors of 0.011 and 0.010. It is 0.001 more negative than published in each of that table's four columns, while the table's other coefficients match.

Table 3: Blair et al.'s estimates with and without interviewers 7, 37, and 71

Blair et al.	Quantity	All interviews	Without 7, 37, 71
Table 1	Endorsement effect, Kashmiri tanzeems	−0.011 (0.006)	−0.020 (0.006)
Table 1	Endorsement effect, Afghan Taliban	−0.015 (0.006)	−0.026 (0.006)
Table 1	Endorsement effect, Al-Qaeda	−0.010 (0.005)	−0.018 (0.005)
Table 1	Endorsement effect, Sectarian tanzeems	−0.008 (0.005)	−0.011 (0.005)
Table 2, col. 1	Effect, middle income	−0.001 (0.006)	−0.013 (0.005)
Table 2, col. 1	Effect, low income	−0.019 (0.008)	−0.021 (0.008)
Table 2, col. 1	Gap: low minus middle income	−0.018 (0.010)	−0.008 (0.010)
Table 2, col. 3	Effect, middle income	0.001 (0.005)	−0.009 (0.005)
Table 2, col. 3	Effect, low income	−0.022 (0.008)	−0.023 (0.008)
Table 2, col. 3	Gap: low minus middle income	−0.023 (0.009)	−0.013 (0.009)
Table 3	Gap: low- minus middle-income district	−0.016 (0.011)	−0.006 (0.010)
Table 4, col. 3	Effect, urban middle income	0.019 (0.009)	0.000 (0.009)
Table 4, col. 3	Effect, urban low income	−0.042 (0.013)	−0.040 (0.013)
Table 4, col. 3	Gap: low minus middle income, urban minus rural	−0.055 (0.020)	−0.037 (0.019)
Table 5, col. 1	Gap: low minus middle income, urban, violent district (incidents)	−0.052 (0.015)	−0.038 (0.014)
Table 5, col. 4	Gap: low minus middle income, urban, violent district (casualties)	−0.061 (0.015)	−0.048 (0.015)

Estimates on Blair et al.'s outcome scale, the mean of a respondent's four answers divided by five, with standard errors in parentheses, from their weighted regressions with sampling points as clusters and provinces as strata; multiply by 125 for points on a 0–100 scale. Table and column numbers refer to Blair et al. (2013a). Columns 1 and 3 of their Table 2 are the specifications with province fixed effects alone and with demographic and attitudinal controls added; Tables 4 and 5 use the latter. "Effect" is the effect of a militant endorsement in the named group; "Gap" is the difference between the effects in two groups. The all-interview estimates reproduce the published values. Table 2, column 1, has 5,067 respondents with all interviews and 4,759 without the three.

from −0.023 to −0.013 (0.009). Across their specifications the gap shrinks by 42% to 55%, and none leaves it more than 1.5 standard errors from zero. The reason is where the three interviewers' respondents sit. Of their 308 respondents with earnings data, 99% are middle or high income, so their endorsed interviews, which record more support than their unendorsed ones, pull the estimated reaction of the middle class toward zero and leave the poor's untouched. A comparison between the effects in two groups of respondents is exposed to any interviewer whose respondents fall mostly in one of them.

The same mechanism runs through their other findings. In their Table 4, the stronger reaction among the urban poor is a three-way interaction of the cue with low income and urban residence: −0.055 (0.020) with all interviews and −0.037 (0.019) without the three, smaller by 33%. The urban poor's own effect is unchanged (−0.042 and −0.040). What changes is the urban middle class, whose effect with all interviews is 0.019 (0.009), a militant endorsement raising support, and without the three is 0.000. In their Table 5, the cue-by-low-income interaction among urban respondents in districts with militant violence shrinks by 22% to 26% and remains more than two standard errors from zero (−0.052 to −0.038, standard error 0.014, in districts with incidents). Their finding that district-level poverty does not change the effect is unchanged (−0.016 to −0.006). And the group-specific effects in their Table 1 grow: with all interviews an endorsement costs a policy 0.9 to 1.9 points depending on the group, and without the three 1.4 to 3.2.

Three interviewers chosen at random do not do this. Dropping 1,000 random triples of interviewers and re-estimating the Table 2 gap gives values between −0.021 and −0.014 in 90% of

draws. The gap comes as close to zero as it does without interviewers 7, 37, and 71 in 1 of the 1,000 draws, and the Table 4 interaction moves as far in none. The three were identified by their answer patterns, before any income result was examined; the estimates with all interviews stay alongside in Table 3 for readers who weigh the evidence differently.

6 Data Quality beyond the Three

The three interviewers' answers to the rest of the questionnaire are uniform but not copied: no two of their interviews match on every question. The survey has a broader problem. For each respondent, the share of questions on which the most similar other respondent in the same sampling point gave identical answers is a standard check for duplicated or fabricated interviews (Kuriakose and Robbins, 2016). Across the 92 closed-ended questions asked of at least half the respondents, the median of that best match is 79%; 31% of respondents have a match on at least 85% of questions, 17% on at least 90%, and 1.8% have an exact twin. The 79 identical pairs come from 13 interviewers, most from interviewers 45, 36, 55 (21, 18, 11 pairs).

How high best matches run depends on how many questions there are, how many answer options they offer, and how alike respondents are (Simmons et al., 2016). Most of these questions are binary, and respondents in a sampling point are neighbors. As a benchmark, I drew 200 sets of 92 questions from the face-to-face interviews of the 2012 American National Election Study, matching the number of answer options of the Pakistan questions; the sets agree by chance slightly more often than the Pakistan questions do.[2] Within the study's primary sampling units, which average about 16 face-to-face respondents, the median best match is about 70%, about 0.1% of respondents match another on at least 85% of questions, and no set produces an exact twin. Pew's 2009 survey of Pakistan, fielded in the same year by a different firm and recording interviewer identifiers, is a closer benchmark (Pew Research Center, 2009). Across its 142 closed-ended questions, the median best match among its 1,254 Pakistani respondents is 69%, 3% match another respondent on at least 85% of questions, and none has an exact twin. Pew asked more questions, which lowers best matches, but each Pew respondent is compared with the whole national sample rather than with eleven neighbors, which raises them. Near-duplicates are far more common in Blair et al.'s survey than in either benchmark: 31% of respondents match a neighbor on at least 85% of questions, against 3% in Pew's survey and 0.1% in the American study, and exact twins occur in neither benchmark. Excluding the duplicating interviewers along with the three leaves the average effect at -0.023 (Table 2).

7 Within-Respondent Patterns

The rest of the analysis excludes the three interviewers, leaving 5,166 respondents and 72 interviewers. Two features of a respondent's four answers bear on whether reactions are carried across

[2] Questions were drawn from the 514 interviewer-administered items asked of at least half of the 2,054 face-to-face respondents, with don't know and refusal kept as answers. Because the study has only 25 binary items, binary items were drawn with replacement, which raises matches. Scripts for both benchmarks are in `exploration/duplicate-benchmarks`; neither survey's data are redistributed.

questions: how answers to different policies move together, and where each respondent's average answer falls.

If reactions were independent across a respondent's questions, the correlation between answers to two policies would be the same with and without endorsements, apart from small changes the mean shift causes on a discrete scale. A reaction carried across all four questions raises the correlations. Table 4 shows 4 of six correlations rising under endorsement, but only 1 of the six changes has an interval that excludes zero. The data are consistent with reactions carried across questions, and they do not establish them.

Table 4: Correlations between policy answers, by condition

Policy pair	Control	Endorsed	Difference	95% interval
Polio – FCR	0.25	0.22	−0.03	[−0.09, +0.04]
Polio – Durand	0.19	0.21	+0.02	[−0.03, +0.08]
Polio – Curriculum	0.16	0.21	+0.06	[−0.01, +0.12]
FCR – Durand	0.55	0.50	−0.05	[−0.10, +0.01]
FCR – Curriculum	0.33	0.39	+0.07	[+0.01, +0.12]
Durand – Curriculum	0.33	0.37	+0.03	[−0.02, +0.08]

Pearson correlations between rescaled answers to two policies, across respondents within each condition, excluding interviewers 7, 37, and 71 and respondents with both control and endorsed items. Intervals are 95% bootstrap intervals for the difference, resampling interviewers.

Table 5 shows the distribution of each respondent's average answer. Indifference predicts no change. Offsetting reactions that are large and common enough empty the middle of the scale in both directions. Opposition alone moves respondents down, and because respondents arriving from above replace those who leave, it leaves the share at the middle roughly unchanged. Without the three interviewers, the share averaging the middle changes by +0.5 [−1.1, +1.8] points and the share one step above by +0.5 [−0.7, +1.9]. The clear gains are below: +2.2 [+1.1, +3.4] points one step below the middle and +1.6 [+0.4, +2.8] points further down. These are net changes between randomized groups, so they cannot show that no one moved up, only that the net movement out of the middle in both directions seen with all interviews is gone.

Table 6 compares these changes with endorsed conditions simulated from the respondents who saw no endorsements. In each simulated condition, a reacting respondent moves each answer one step in their direction with a fixed probability, and the share of opponents is set so the expected average effect equals the observed one. Opposition alone produces changes at the middle of +0.5 to +0.8 points and one step above of +0.3 to +1.1 points, in line with what is observed. Simulated conditions in which up to 30% react favorably produce changes at the middle of +0.3 to +0.6 points, which also fit. The observed rise in spread fits both kinds of condition. The observed gain one step below the middle is larger than the average gain in any simulated condition (at most +1.5 points), so neither kind reproduces it fully. Without the three interviewers, the comparison does not require favorable reactions, and it cannot rule them out.

Table 5: Distribution of each respondent's average answer, by condition

Mean answer	Control	Endorsed	Difference (pp)	95% interval
$\leq 6/16$	3.5%	5.1%	+1.6	[+0.4, +2.8]
7/16	2.9%	5.1%	+2.2	[+1.1, +3.4]
8/16	6.6%	7.0%	+0.5	[−1.1, +1.8]
9/16	7.5%	8.0%	+0.5	[−0.7, +1.9]
10/16	12.6%	12.5%	−0.1	[−1.9, +1.8]
11/16–13/16	34.1%	31.6%	−2.4	[−5.3, +0.2]
14/16–15/16	15.8%	14.4%	−1.4	[−3.0, 0.0]
16/16	17.2%	16.3%	−0.9	[−2.7, +0.9]

Average of a respondent's four answers on the 0–1 scale, which moves in steps of 1/16; the middle of the scale is 8/16. Sparse bins are pooled. Excludes interviewers 7, 37, and 71. Differences are in percentage points of respondents; intervals are 95% bootstrap intervals resampling interviewers.

Table 6: Observed and simulated changes under endorsement

Endorsed arm	Oppose	Favor	Answers moved	Spread (×100)	Midpoint (pp)	One below (pp)	One above (pp)
Observed				+0.5 [+0.1, +0.8]	+0.5 [−1.1, +1.8]	+2.2 [+1.1, +3.4]	+0.5 [−0.7, +1.9]
Opposition only	35%	0%	25%	+0.1 [−0.1, +0.4]	+0.7 [−0.8, +2.1]	+1.0 [0.0, +1.9]	+1.1 [−0.3, +2.5]
Opposition only	17%	0%	50%	+0.2 [−0.1, +0.5]	+0.8 [−0.5, +2.1]	+1.0 [0.0, +2.0]	+0.7 [−0.8, +2.1]
Opposition only	9%	0%	100%	+0.4 [+0.1, +0.7]	+0.5 [−0.8, +1.9]	+0.8 [−0.1, +1.8]	+0.3 [−1.2, +1.7]
Both directions	23%	10%	50%	+0.3 [0.0, +0.6]	+0.6 [−0.8, +1.9]	+1.1 [+0.1, +2.1]	+0.7 [−0.8, +2.0]
Both directions	30%	20%	50%	+0.4 [+0.1, +0.6]	+0.4 [−0.9, +1.9]	+1.3 [+0.3, +2.4]	+0.5 [−0.9, +2.1]
Both directions	36%	30%	50%	+0.4 [+0.2, +0.7]	+0.3 [−1.3, +1.5]	+1.5 [+0.5, +2.6]	+0.5 [−0.9, +2.2]

Changes from the control condition, excluding interviewers 7, 37, and 71. The first row is observed, with 95% bootstrap intervals resampling interviewers. Other rows are simulated endorsed conditions built by resampling control respondents to the size of the endorsed condition. A reacting respondent moves each answer one step (0.25) down if opposed or up if favorable, with the probability shown as answers moved, truncated at the ends of the scale. The opposed share is set so the expected average effect equals the observed one. Simulated rows report the mean and the 2.5th and 97.5th percentiles over 500 simulated samples. Spread is the variance of a respondent's average answer. Midpoint, one below, and one above are the shares whose answers average 8/16, 7/16, and 9/16.

8 How Many React Favorably?

The within-respondent patterns do not say how many respondents react favorably. Each route to that number rests on assumptions the data cannot check.

8.1 The Measurement Model

The measurement model of Bullock et al. (2011) relates ordinal policy answers to a latent policy position x_i for each respondent and a reaction s_{ijk} to group k's endorsement of question j, which shifts the position before it is mapped to an answer through question-specific loadings and cutpoints. In the baseline specification,

$$s_{ijk} \sim \mathcal{N}(\lambda_k, \omega_k^2), \qquad P(s_{ijk} > 0) = \Phi(\lambda_k/\omega_k). \tag{2}$$

Blair et al. (2014) proposed reading the probability of a positive reaction as the share of respondents who support the endorser, and Rosenfeld et al. (2016) treat it as the model's main quantity of interest. The mean and variance of the reaction are common across policies, and each reaction is drawn independently across a respondent's questions. A positive latent reaction need not move an answer into a higher category, so $P(s_{ijk} > 0)$ is not the share of answers an endorsement raises.

I fit the baseline model with `endorse`, with and without the three interviewers.[3] Without them, the probability of a positive latent reaction, averaged equally over the four groups, is 38% (95% credible interval 29%–45%); with all interviews it is 43%. Table 7 reports the estimates by group. The Afghan Taliban have the lowest estimate, 27% (4%–41%). For comparison, in Pew's 2009 survey of Pakistan, 10% of respondents said they viewed the Taliban favorably and 20% gave no answer (Pew Research Center, 2009), though a stated opinion and a latent reaction to an endorsement are different quantities. The model's estimate of how many react favorably comes from its normal reaction distribution and its independence assumption, not from features of the data checked here. With a mean reaction this close to zero (Table 7), the normal form does most of the work: whatever the spread, close to half of a normal distribution centered near zero lies above zero, and Section 7 shows that a population in which no one reacts favorably fits the data as well. The quantity is not always near one half. In Afghanistan, Blair et al. (2014) estimate 17.5% support for international forces from an endorsement experiment and 16% from a list experiment on the same respondents, where the mean reaction is clearly negative; in a validation against a referendum, Rosenfeld et al. (2016) estimate 52% from an endorsement experiment against a 65% vote. Its reading as a share is weakest where the mean reaction is small, as here.

[3] Each fit uses three chains with starting values dispersed across negative, zero, and positive mean reactions. Each chain runs for 200,000 iterations, discards 20,000, and retains every 40th draw. The largest Gelman–Rubin statistic across the mean and variance parameters in either fit is 1.04, and the smallest effective sample size across mean reactions is 1,389. The fits use the package's default settings. For this specification, `endorse` 1.6.2 does not apply its documented scaled inverse-chi-squared prior to the reaction variance; its code draws the variance under a flat prior on $(0, 10]$, and changing the documented prior's arguments leaves the draws unchanged. The posterior reaction variances lie well below 1, so the truncation does not bind, and the share of positive reactions is set by the data and the model's form, not by a prior.

Table 7: Positive latent reactions under the baseline measurement model, excluding the three interviewers

Endorser	$P(s > 0)$	95% CrI	Mean reaction (ideal-point SDs)	Reaction SD (ideal-point SDs)
Kashmiri tanzeems	37%	[23%, 47%]	−0.11	0.36
Afghan Taliban	27%	[4%, 41%]	−0.16	0.29
Al-Qaeda	45%	[38%, 50%]	−0.08	0.57
Sectarian tanzeems	42%	[29%, 51%]	−0.07	0.36
Average	38%	[29%, 45%]	−0.10	0.39

Unweighted sample excluding interviewers 7, 37, and 71, $n = 5,166$. CrI denotes a 95% credible interval. Mean reaction and reaction SD are posterior means in ideal-point SD units. The average row weights the four endorsers equally.

8.2 Bounds and Restrictions

Without a model, the experiment identifies the distributions of answers with and without endorsements but not how they pair within a person. Allowing every pairing bounds the fraction of answers an endorsement raises between 0% and 59%, the sharp bounds for an ordinal outcome of Lu et al. (2018).

Fixing the sizes of favorable and unfavorable effects in Equation (1) turns the average effect into a share. Table 8 collects several such restrictions. The implied shares run from 15% to 98%. The response ceiling widens the range: 41% of answers without endorsements are at the top of the scale, where a favorable reaction cannot show. If favorable and unfavorable respondents both move one category, truncation at the top implies a favorable share of 57% despite a negative mean.

Table 8: Favorable shares implied by restrictions on effect sizes

Restriction	Assumption	Implied fraction
Extreme opponent reaction	$\delta^- = -1$, $\delta^+ = 0$	98%
Binary switching	All supporters $\to 1$, opponents $\to 0$	73%
5-point scale, 2-step shift	Shift ± 2 categories, truncated	65%
5-point scale, 1-step shift	Shift ± 1 category, truncated	57%
Normal heterogeneity ($\sigma = 0.05$)	$\delta_i \sim \mathcal{N}(\beta, 0.05^2)$	34%
Symmetric effects ($d = 0.05$)	$\delta^+ = +0.05$, $\delta^- = -0.05$	29%
Symmetric effects ($d = 0.03$)	$\delta^+ = +0.03$, $\delta^- = -0.03$	15%

All rows use the average effect excluding interviewers 7, 37, and 71, −0.021, and, where required, the control mean 0.746. Fractions refer to the class defined by each restriction. Binary switching is contradicted by the observed interior categories. The normal row is a continuous approximation to individual effects on the observed scale; the extreme row is an unaffected fraction. Category shifts assign both classes the control distribution and truncate at the ends of the scale.

9 Discussion

Interviewer effects are a standard concern in survey research (West and Blom, 2017), and they bear on endorsement experiments in two ways. First, randomizing the questionnaire version within

interviewers does not protect the treatment effect from them. An interviewer who records a fixed answer string under each version adds a fixed difference to the estimated effect, and here the three interviewers' differences had the opposite sign from the pooled effect and were among the largest in the survey. Second, a comparison of effects between groups of respondents is exposed to any interviewer whose respondents fall mostly in one group; here the three interviewers' respondents were almost all middle or high income, and the poverty gradient went with them. Third, analyses that use the joint distribution of a respondent's answers are especially exposed, because a repeated answer string is a joint pattern. Intervals that resample respondents treat an interviewer's repeated strings as independent evidence; resampling interviewers does not, and on its own it makes the published midpoint pattern indistinguishable from zero. A test of whether the effect varies across interviewers more than chance allows needs nothing beyond the identifiers, and it finds the three without being told where to look.

The checks used here come from work on detecting fabricated and shortcut interviews. Interviewers who invent answers tend to record answer strings that vary too little within an interview and are too alike across interviews (Schräpler and Wagner, 2005; Menold and Kemper, 2014; Blasius and Thiessen, 2015), and screens built on such indicators can single out falsifiers from the data alone (Bredl et al., 2012; Schwanhäuser et al., 2022; Olbrich et al., 2024). Survey organizations pair these screens with timestamps, callbacks, and reinterviews (Murphy et al., 2016; DeMatteis et al., 2020). Percent match applies the logic to whole interviews (Kuriakose and Robbins, 2016), and its cutoffs depend on the questionnaire and the sample (Simmons et al., 2016). The same idea, that invented data are too regular, has exposed teachers who altered answer sheets (Jacob and Levitt, 2003) and fabricated experimental data (Simonsohn, 2013; Broockman et al., 2015). Such checks locate problems; they do not establish fabrication. Here the evidence stops short of it: the three interviewers' answers to the rest of the questionnaire are uniform but not copied. What the checks establish is narrower and enough for the argument: the patterns that suggested offsetting reactions depend on three interviewers.

The favorable share needs a different design. Randomizing whether the group is said to support or oppose the policy separates reactions to the group from a general push away from the middle of the scale: a respondent who favors the group should move in opposite directions under the two cues. Policies with intermediate baseline support, or a finer response scale, leave room for favorable reactions to appear; here 41% of control answers are already at the top. Asking the policy questions without endorsements in an earlier wave would observe each respondent's own shift. Repeated-measures designs in survey experiments have generally not altered treatment effects (Clifford et al., 2021), although consistency pressure across waves would need to be checked.

10 Conclusion

Three of 75 interviewers in Blair et al.'s survey recorded the same answer strings in most of their interviews, and their interviews make middle- and high-income Pakistanis look warmer toward militant groups than the rest of the sample is. Without them, the poor and the middle class react to militant endorsements about alike, and the gap between them that is Blair et al.'s central

estimate is not distinguishable from zero; the larger reactions they report among the urban poor and in violent districts shrink by a fifth to a third and remain. The evidence for low regard is the small average effect, and it stays small: the cost of a militant endorsement grows from 1.3 to 2.1 points on a 0–100 scale. The three interviewers also produce the appearance that endorsements push respondents out of the middle of the scale in both directions; without them that appearance goes away, and what remains fits opposition alone. The standard model puts the share who react favorably at 38%, but with a mean reaction near zero that number comes from the model's normal reaction distribution, and a population in which no one reacts favorably fits these data as well.

References

Abadie, A., Athey, S., Imbens, G. W., and Wooldridge, J. M. (2023). When should you adjust standard errors for clustering? *Quarterly Journal of Economics*, 138(1):1–35.

Blair, G., Fair, C. C., Malhotra, N., and Shapiro, J. N. (2013a). Poverty and support for militant politics: Evidence from Pakistan. *American Journal of Political Science*, 57(1):30–48.

Blair, G., Fair, C. C., Malhotra, N., and Shapiro, J. N. (2013b). Replication data for: Poverty and support for militant politics: Evidence from Pakistan. Harvard Dataverse. File `fms09-data-weighted-20june.dta`.

Blair, G., Imai, K., and Lyall, J. (2014). Comparing and combining list and endorsement experiments: Evidence from Afghanistan. *American Journal of Political Science*, 58(4):1043–1063.

Blasius, J. and Thiessen, V. (2015). Should we trust survey data? assessing response simplification and data fabrication. *Social Science Research*, 52:479–493.

Bredl, S., Winker, P., and Kötschau, K. (2012). A statistical approach to detect interviewer falsification of survey data. *Survey Methodology*, 38(1):1–10.

Broockman, D., Kalla, J., and Aronow, P. (2015). Irregularities in LaCour (2014). Unpublished report, Stanford University, University of California, Berkeley, and Yale University.

Bullock, W., Imai, K., and Shapiro, J. N. (2011). Statistical analysis of endorsement experiments: Measuring support for militant groups in Pakistan. *Political Analysis*, 19(4):363–384.

Clifford, S., Sheagley, G., and Piston, S. (2021). Increasing precision without altering treatment effects: Repeated measures designs in survey experiments. *American Political Science Review*, 115(3):1048–1065.

DeMatteis, J. M., Young, L. J., Dahlhamer, J., Langley, R. E., Murphy, J., Olson, K., and Sharma, S. (2020). Falsification in surveys. Report of the aapor/asa task force on data falsification, American Association for Public Opinion Research and American Statistical Association.

Ding, P., Feller, A., and Miratrix, L. (2016). Randomization inference for treatment effect variation. *Journal of the Royal Statistical Society: Series B*, 78(3):655–671.

Jacob, B. A. and Levitt, S. D. (2003). Rotten apples: An investigation of the prevalence and predictors of teacher cheating. *The Quarterly Journal of Economics*, 118(3):843–877.

Kuriakose, N. and Robbins, M. (2016). Don't get duped: Fraud through duplication in public opinion surveys. *Statistical Journal of the IAOS*, 32(3):283–291.

Lu, J., Ding, P., and Dasgupta, T. (2018). Treatment effects on ordinal outcomes: Causal estimands and sharp bounds. *Journal of Educational and Behavioral Statistics*, 43(5):540–567.

Lyall, J., Blair, G., and Imai, K. (2013). Explaining support for combatants during wartime: A survey experiment in Afghanistan. *American Political Science Review*, 107(4):679–705.

Manski, C. F. (1990). Nonparametric bounds on treatment effects. *American Economic Review*, 80(2):319–323.

Menold, N. and Kemper, C. J. (2014). How do real and falsified data differ? psychology of survey response as a source of falsification indicators in face-to-face surveys. *International Journal of Public Opinion Research*, 26(1):41–65.

Murphy, J., Biemer, P., Stringer, C., Thissen, R., Day, O., and Hsieh, Y. P. (2016). Interviewer falsification: Current and best practices for prevention, detection, and mitigation. *Statistical Journal of the IAOS*, 32(3):313–326.

Nicholson, S. P. (2012). Polarizing cues. *American Journal of Political Science*, 56(1):52–66.

Olbrich, L., Kosyakova, Y., Sakshaug, J. W., and Schwanhäuser, S. (2024). Detecting interviewer fraud using multilevel models. *Journal of Survey Statistics and Methodology*, 12(1):14–35.

Pew Research Center (2009). Spring 2009 Pew global attitudes survey. Dataset. Pakistan sample; weighted shares computed from the respondent-level file.

Rosenfeld, B., Imai, K., and Shapiro, J. N. (2016). An empirical validation study of popular survey methodologies for sensitive questions. *American Journal of Political Science*, 60(3):783–802.

Schräpler, J.-P. and Wagner, G. G. (2005). Characteristics and impact of faked interviews in surveys: An analysis of genuine fakes in the raw data of SOEP. *Allgemeines Statistisches Archiv*, 89(1):7–20.

Schwanhäuser, S., Sakshaug, J. W., and Kosyakova, Y. (2022). How to catch a falsifier: Comparison of statistical detection methods for interviewer falsification. *Public Opinion Quarterly*, 86(1):51–81.

Shiraito, Y. and Imai, K. (2022). *endorse: Bayesian Measurement Models for Analyzing Endorsement Experiments*. R package version 1.6.2.

Simmons, K., Mercer, A., Schwarzer, S., and Kennedy, C. (2016). Evaluating a new proposal for detecting data falsification in surveys. *Statistical Journal of the IAOS*, 32(3):327–338.

Simonsohn, U. (2013). Just post it: The lesson from two cases of fabricated data detected by statistics alone. *Psychological Science*, 24(10):1875–1888.

West, B. T. and Blom, A. G. (2017). Explaining interviewer effects: A research synthesis. *Journal of Survey Statistics and Methodology*, 5(2):175–211.